CS-663:
Depth Estimation and Refocusing

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Outline

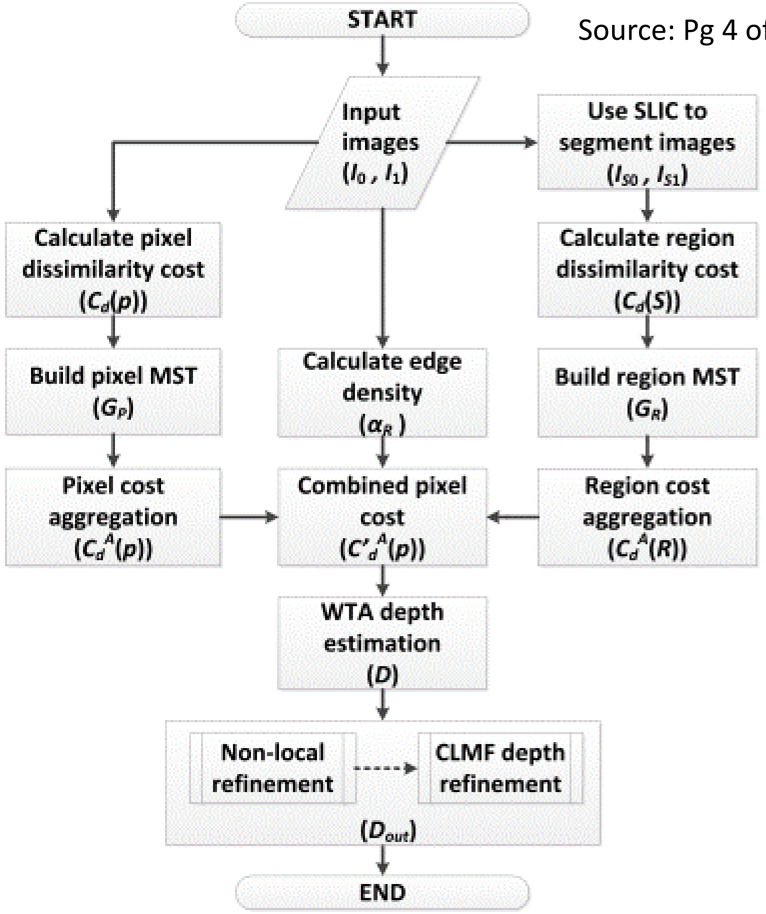
I. Context
II. Depth Estimation
    i. Pixel level Dissimilarity Cost
    ii. Region level Dissimilarity Cost
    iii. Kruskal’s Algorithm
    iv. Adaptive Fusion of costs
    v. Non-Local Disparity Map Refinement
    vi. Cross-Based Local Multi-point Filtering
III. Post Capture Refocusing

Context

- Stereo Images are the pair of images of a scene as seen by 2 slightly separated cameras (eg. biocular humans)
- Responsible for perception of depth by humans

Depth Estimation

- Pixel Level for fine aggregation
  - Useful for edges and fine textures
- Region level for coarse aggregation
  - Useful for uniform region
- Use of MST for Non Local Cost Aggregation
- Adaptive Fusion of 2 levels of cost
- Non- Local Refinement



Source: Pg 4 of [1]

Pixel Level Dissimilarity Cost

$$C_d(p) = \beta e_i(p, p_d) + (1 - \beta) e_g(p, p_d),$$

$$e_i(p, p_d) = \min\left(|I_0(p) - I_1(p_d)|, T_i\right),$$

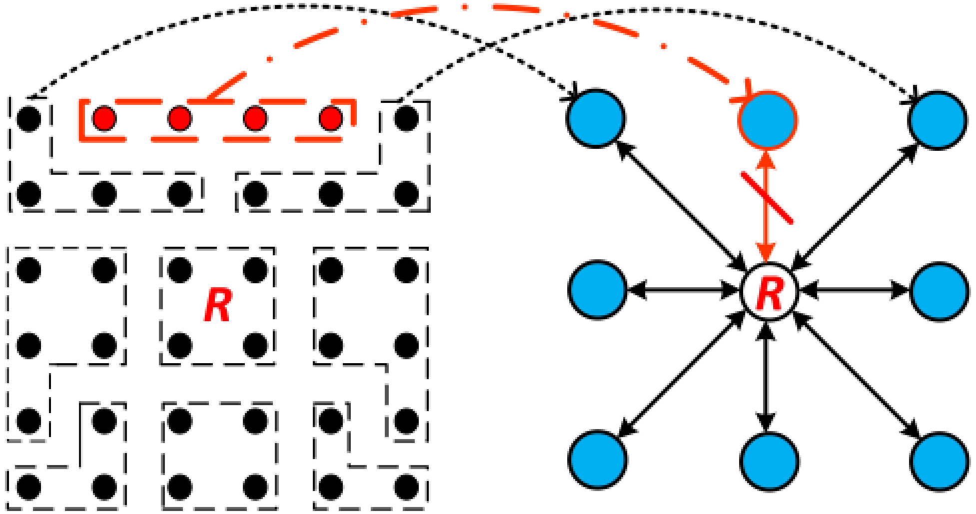
$$e_g(p, p_d) = \min\left(|I_0'(p) - I_1'(p_d)|, T_g\right),$$

[4] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Susstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," IEEE Trans. Pattern Anal. Mach. Intell., vol. 34, no. 11, pp. 2274–2282, Nov. 2012

Region Level Dissimilarity Cost

- Regions defined by superpixels (by SLIC - Super Linear Iterative Clustering - algorithm) [4]
- Edge Weight $\omega_R(S,T) = |I_S - I_T|$ Max (across channels) of Abs. Diff
- Cost in region is the sum of pixel-level cost of all pixels in region

Source: Pg 5 of [1]



Minimum Spanning Tree

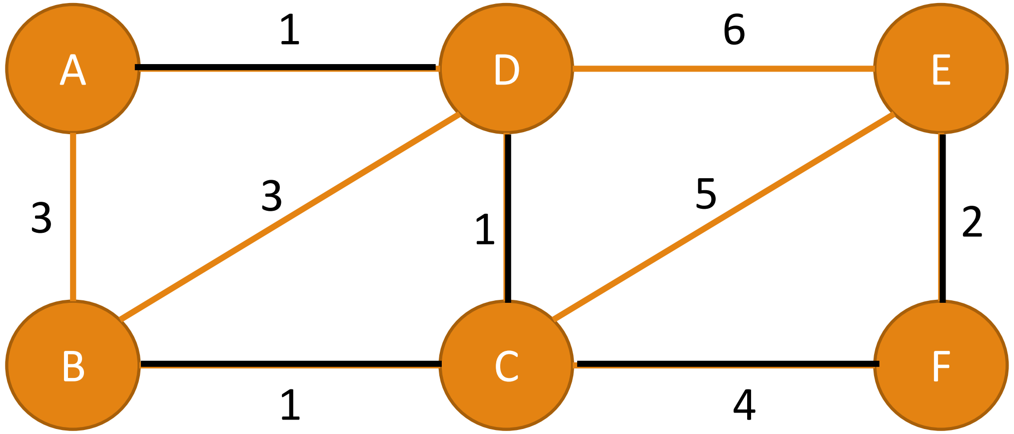
- Non Local Cost Aggregation
  a. No implicit local smoothness assumption
  b. Fast, yet no discretization (unlike fast-Bilateral Filtering)
  c. More natural similarity metric: No user-specified window

- Requires a single forward and backward pass of tree
  - Low Computational Complexity (2+, 3×)

Source: Tushar Roy – Coding Made Simple
https://www.youtube.com/watch?v=fAuF0EuZVCk

Kruskal's Algorithm

- Obtains the minimum spanning tree from weighted undirected graph

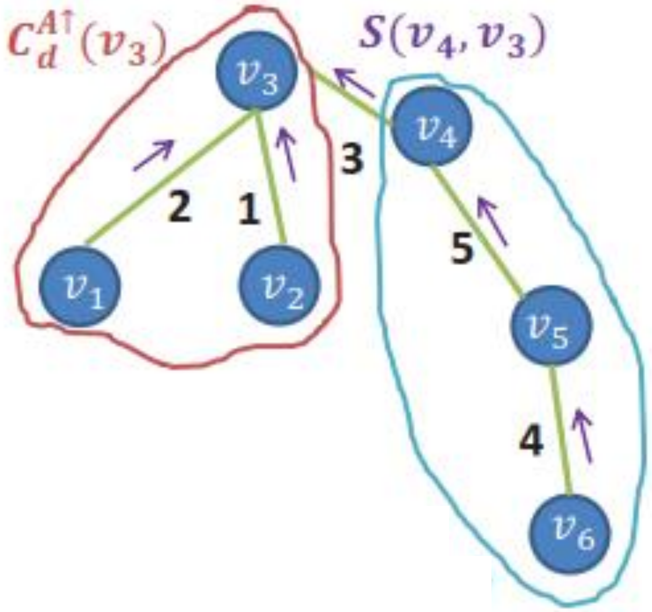


AD – 1 ✓
BC – 1 ✓
CD – 1 ✓
EF – 2 ✓
BD – 3 ✗
AB – 3 ✗
CF – 4 ✓
CE – 5 ✗
DE – 6 ✗

Cost Aggregation

$$\mathcal{S}(p,q) = \mathcal{S}(q,p) = \exp\left(-\mathcal{D}(p,q)/\sigma\right)$$

$$C_d^A(p) = \sum_q S(p,q)C_d(q)$$



$$C_d^{A\uparrow}(v) = C_d(v) + \sum_{P(v_c)=v} \mathcal{S}(v, v_c).C_d^{A\uparrow}(v_c)$$

$$C_d^A(v) = \mathcal{S}(v, P(v)).C_d^A(P(v)) + [1 - \mathcal{S}^2(v, P(v))].C_d^{A\uparrow}(v)$$

Source: Pg 4 of [2]

Adaptive Fusion of Costs

- Pixel level cost accurate near colour/depth discontinuities
- At texture-less regions, region cost is more accurate
- Edges are detected by Canny Edge Detection

$$C_d^{A\prime}(p) = \alpha_R C_d^A(p) + (1 - \alpha_R) C_d^A(R)$$

$$\alpha_R = \frac{N_e}{N_R}$$

Non Local Refinement

- 2 Step Refinement
- Fix for occlusion: Mutual Consistency Check
- Fix for large texture-less regions: CLMF

Mutual Consistency Check

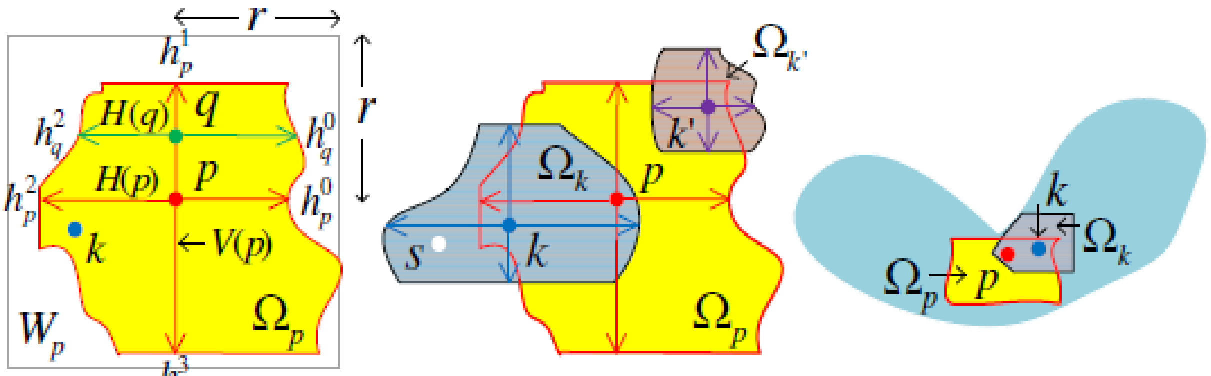
- Disparity maps of left and right image should be consistent
- Points of consistency are termed ‘stable’

$$C_d^{new}(p) = \begin{cases} |d - D(p)| & p \text{ is stable and } D(p) > 0, \\ 0 & \text{else.} \end{cases}$$

- Aggregate ‘Consistent Cost’ using pixel-level MST

Cross based Local Multipoint Filtering

- Edge preserving smoothing - local multipoint vs. single point estimates of BF
- Uses cross based local support decision and integration technique to perform zero order polynomial modeling over pointwise adaptive support regions.
- Memory efficient and faster compared to Bilateral filtering.



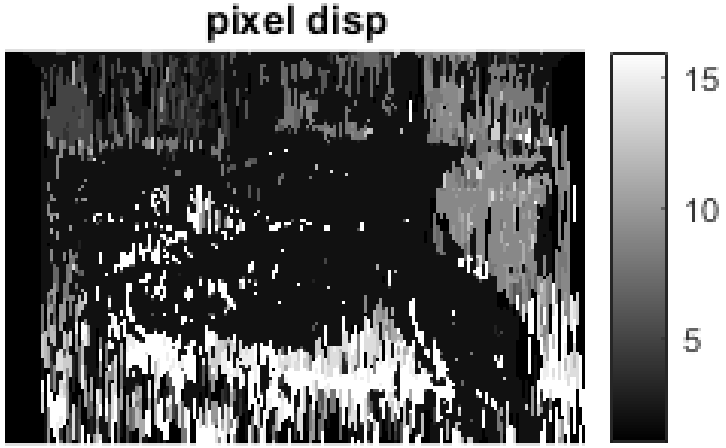
(a) (b) (c)

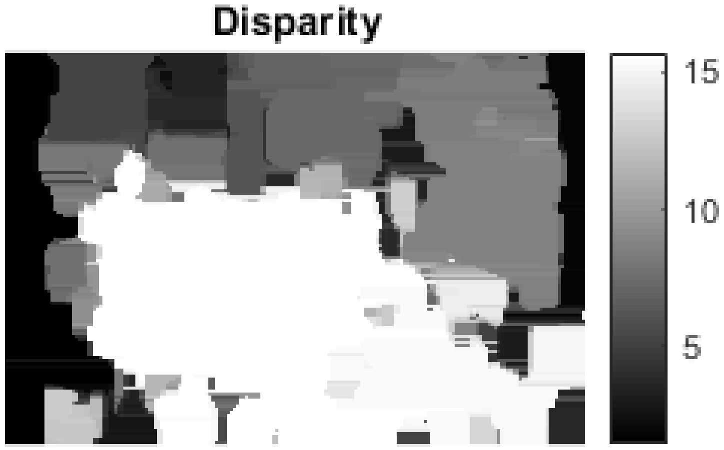
a) Smoothened image vs Original image
Threshold = 1/255

b) Smoothened image vs Original image
Threshold = 10/255

Sanity Check: Constant Shift

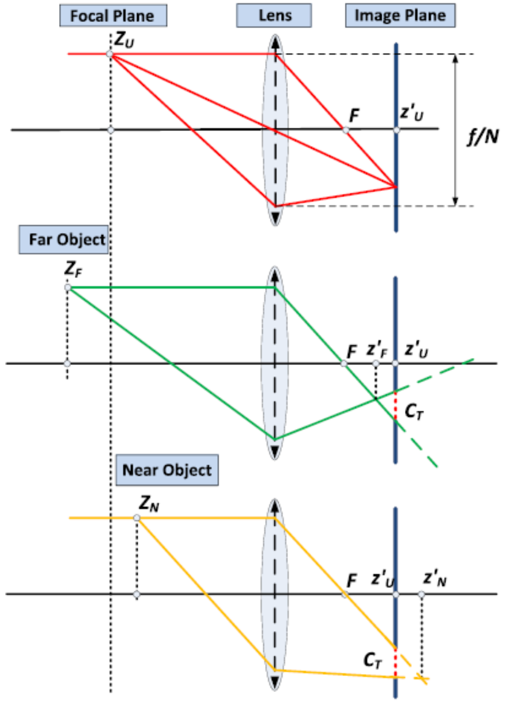
Depth Estimation Result





Depth based Refocusing

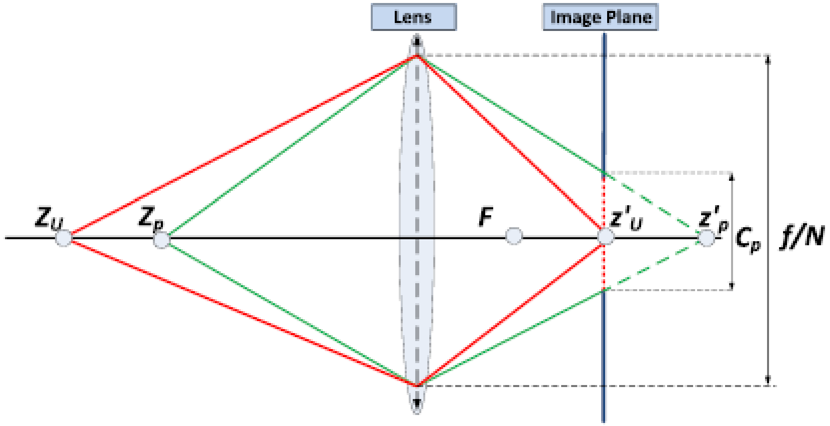
- User will initially scribble along all points required to be in focus. All points lying outside the planes selected will be out of focus by gaussian blurring them proportional to the radius of their circle of confusion .



Point in focus
Radius of circle of confusion = 0

Points out of focus
Radius of circle of confusion is finite. More out of focus larger the circle of confusion.

Depth based Refocusing

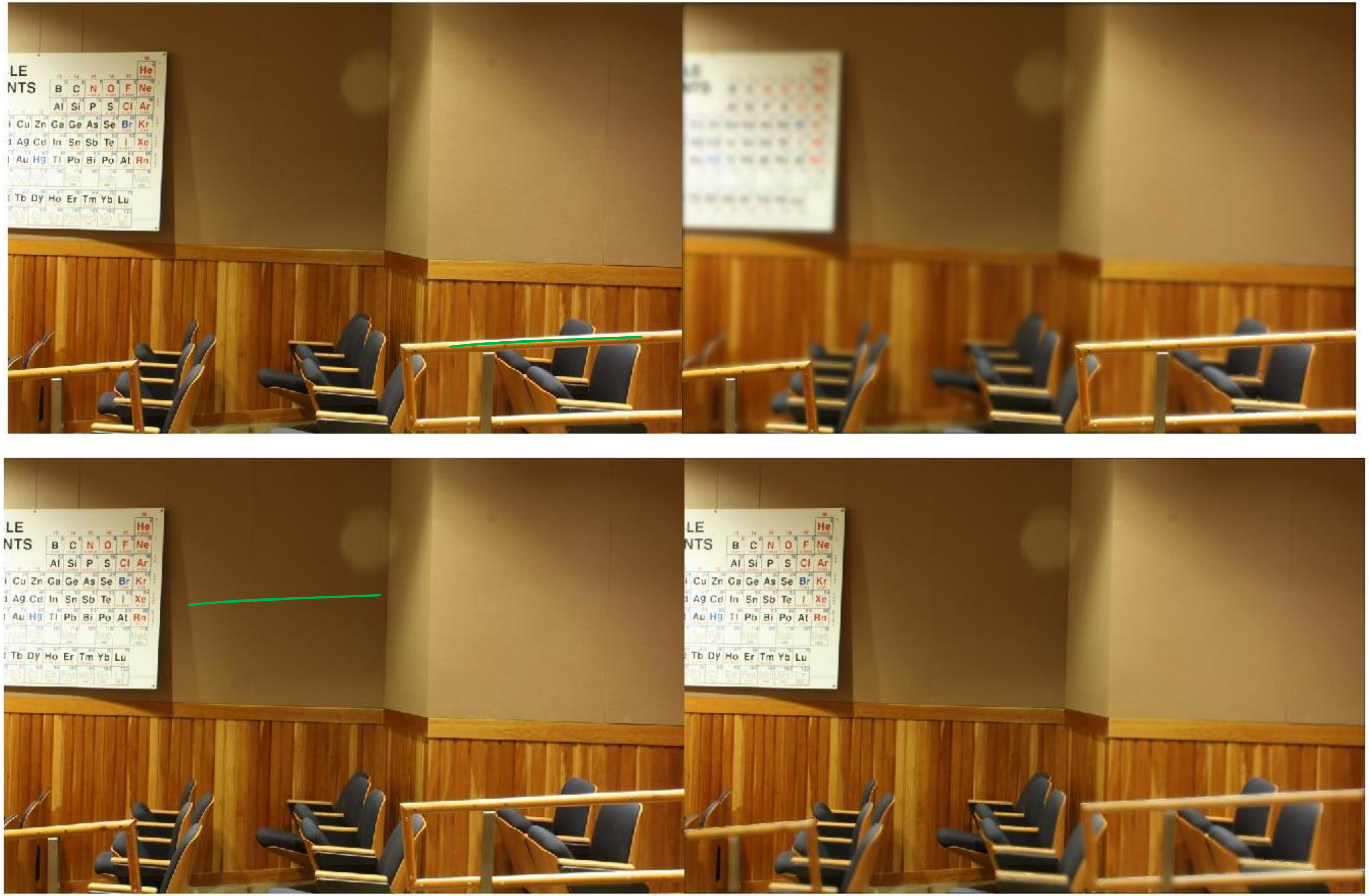


$$Z_U = \frac{1}{3}(Z_F - Z_N) + Z_N.$$

$$\frac{C_p}{f/N} = \frac{z'_p - z'_U}{z'_p}$$

$$\sigma_p = K * \frac{C_p}{p_s},$$

$$G_p(u,v) = \frac{1}{2{\sigma_p}^2} \exp\left(\frac{-(u^2+v^2)}{2{\sigma_p}^2}\right) \cdot \delta\left(D_{out}(p+(u,v))\right),$$



*Tested using Ground truth disparity image

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Bibliography

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[2] Q. Yang. A non-local cost aggregation method for stereo matching. In 2012 IEEE Conference on Computer Vision and Pattern Recognition , pages 1402–1409, June 2012.

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[4] R. Achanta, A. Shaji, K. Smith, A. Lucchi, P. Fua, and S. Susstrunk, "SLIC superpixels compared to state-of-the-art superpixel methods," IEEE Trans. Pattern Anal. Mach. Intell., vol. 34, no. 11, pp. 2274–2282, Nov. 2012

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